CALPHADHEACalc: Governing Equations & Variable Reference Manual

Mathematical Formulations, Parameter Definitions, SI Units, and Subsystem Mapping

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Abstract — This reference manual provides an authoritative compendium of all governing mathematical equations, CALPHAD thermodynamic models, lattice distortion mechanics, empirical descriptors, transport phenomena, environmental kinetics, and kinematic crystallography models implemented within the CALPHADHEACalc engineering workstation. Each formulation includes a description of where it is used in the software, followed by complete variable definitions with standard physical dimensions and SI units. All content is presented in standard monochrome academic format without ornamental design elements.

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1. CALPHAD Solution Thermodynamics & Phase Equilibria

Primary Engines: HEAThermoEngine.m, HEATDBEngine.m | Workstation Modules: Tab 1, Tab 2, Iterative Console

1.1 Total Molar Gibbs Free Energy of Phase φ

Where Used: HEAThermoEngine.m. Fundamental thermodynamic objective function minimized during phase equilibrium stepping across temperature and composition.

$$G_m^{\phi}(T, \mathbf{c}) = {}^{\text{ref}}G^{\phi}(T) + {}^{\text{id}}G^{\phi}(T) + {}^{\text{E}}G^{\phi}(T) + {}^{\text{mag}}G^{\phi}(T) \tag{1}$$

Variable Definitions:

- $G_m^{\phi}(T, \mathbf{c})$: Total molar Gibbs free energy of multicomponent phase ϕ at absolute temperature T [$\mathbf{J \cdot mol^{-1}}$].
- T: Absolute thermodynamic temperature [$\mathbf{K}$].
- $\mathbf{c} = (c_1, c_2, \ldots, c_n)$: Vector of mole fractions of the n constituent alloy components [dimensionless, $\sum_{i=1}^{n} c_i = 1$].
- ${}^{\text{ref}}G^{\phi}(T)$: Reference energy of pure elements in the physical state of phase ϕ [$\mathbf{J \cdot mol^{-1}}$].
- ${}^{\text{id}}G^{\phi}(T)$: Ideal configurational entropy of mixing contribution [$\mathbf{J \cdot mol^{-1}}$].
- ${}^{\text{E}}G^{\phi}(T)$: Non-ideal excess Gibbs free energy resulting from interatomic pair interactions [$\mathbf{J \cdot mol^{-1}}$].
- ${}^{\text{mag}}G^{\phi}(T)$: Magnetic ordering Gibbs energy contribution based on the Inden-Hillert-Jarl model [$\mathbf{J \cdot mol^{-1}}$].

1.2 Reference Energy (SGTE Unary Lattice Stabilities)

Where Used: HEATDBEngine.m. Evaluates standard pure element Gibbs free energies for 28 elements across FCC, BCC, HCP, and Liquid states.

$${}^{\text{ref}}G^{\phi}(T) = \sum_{i=1}^{n} c_i \, {}^{\circ}G_i^{\phi}(T) \tag{2}$$

$${}^{\circ}G_i^{\phi}(T) - H_i^{\text{SER}} = a_i + b_i T + c_i T \ln T + d_i T^2 + e_i T^{-1} + f_i T^3 \tag{3}$$

Variable Definitions:

- ${}^{\circ}G_i^{\phi}(T)$: Standard molar Gibbs energy of pure element i in crystal structure ϕ [$\mathbf{J \cdot mol^{-1}}$].
- H_i^{SER}: Enthalpy of pure element i in its Standard Element Reference (SER) state at $\mathbf{298.15\ K}$ and $\mathbf{10^5\ Pa}$ [$\mathbf{J \cdot mol^{-1}}$].
- $a_i, b_i, c_i, d_i, e_i, f_i$: Temperature-dependent polynomial coefficients assessed in the SGTE unary database (Dinsdale, 1991).
- c_i: Mole fraction of element i [$\mathbf{mol/mol}$].

1.3 Ideal Solution Configurational Entropy

Where Used: HEAThermoEngine.m. Evaluates entropic stabilization energy driving single-phase solid solution formation.

$${}^{\text{id}}G^{\phi}(T) = RT \sum_{i=1}^{n} c_i \ln c_i \tag{4}$$

Variable Definitions:

- R: Universal gas constant ($\mathbf{8.314462\ J \cdot mol^{-1} \cdot K^{-1}}$).
- T: Absolute thermodynamic temperature [$\mathbf{K}$].
- c_i: Mole fraction of component i ($\mathbf{0 < c_i \leq 1}$).

1.4 Redlich-Kister Polynomial Excess Gibbs Energy

Where Used: HEAThermoEngine.m. Computes enthalpic and entropic deviations from ideality using assessed binary Redlich-Kister interaction parameters.

$$^{\mathrm{E}}G^{\phi}(T) = \sum_{i=1}^{n-1}\sum_{j=i+1}^{n} c_i c_j \sum_{k=0}^{p} L_{ij}^{\phi,(k)}(T)\,(c_i - c_j)^k, \quad L_{ij}^{\phi,(k)}(T) = a_k + b_k T \tag{5}$$

Variable Definitions:

- $L_{ij}^{\phi,(k)}(T)$: k-th order Redlich-Kister binary interaction parameter between elements i and j in phase ϕ [$\mathbf{J \cdot mol^{-1}}$].
- a_k: Enthalpic interaction coefficient [$\mathbf{J \cdot mol^{-1}}$].
- b_k: Entropic interaction coefficient [$\mathbf{J \cdot mol^{-1} \cdot K^{-1}}$].
- p: Polynomial expansion order ($p = 0$ for regular solution, $p \geq 1$ for sub-regular).

1.5 Chemical Potential Equalization (Multiphase Equilibrium)

Where Used: HEAThermoEngine.m. Core iterative Newton-Raphson solver establishing multiphase equilibrium and mass conservation.

$$\mu_i^{\alpha}(T, \mathbf{x}^{\alpha}) = \mu_i^{\beta}(T, \mathbf{x}^{\beta}) = \cdots = \mu_i^{\nu}(T, \mathbf{x}^{\nu}) \quad (\forall i = 1, \ldots, n) \tag{6}$$

$$\sum_{\phi=1}^{\Phi} f_{\phi} x_i^{\phi} = c_i^0, \quad \sum_{\phi=1}^{\Phi} f_{\phi} = 1 \tag{7}$$

Variable Definitions:

- μ_i^{ϕ}: Partial molar chemical potential of component i in phase ϕ [$\mathbf{J \cdot mol^{-1}}$], defined as $\mu_i^{\phi} = (\partial G^{\phi}/\partial n_i)_{T,P,n_{j \neq i}}$.
- f_{ϕ}: Molar phase fraction of phase ϕ ($0 \leq f_{\phi} \leq 1, \sum f_{\phi} = 1$) [dimensionless].
- $\mathbf{x}^{\phi} = (x_1^{\phi}, \ldots, x_n^{\phi})$: Composition vector of phase ϕ [$\mathbf{mol/mol}$].
- c_i^0: Nominal overall alloy mole fraction of component i [$\mathbf{mol/mol}$].
- $\alpha, \beta, \ldots, \nu$: Active equilibrium phases (FCC_A1, BCC_A2, B2, LIQUID, SIGMA).

2. Empirical HEA Descriptors & Solidification Kinetics

Primary Engine: HEAThermoEngine.m | Workstation Modules: Tab 1, Tab 4, Tab 8, Diagnostics Console

2.1 Miedema Enthalpy of Mixing & Configurational Entropy

Where Used: HEAThermoEngine.m. Evaluates semi-empirical macroscopic atom interaction enthalpies and configurational entropy.

$$\Delta H_{\mathrm{mix}} = \sum_{i=1}^{n-1}\sum_{j=i+1}^{n} 4\,\Omega_{ij}\,c_i\,c_j, \quad \Delta S_{\mathrm{mix}} = -R\sum_{i=1}^{n} c_i \ln c_i \tag{8}$$

Variable Definitions:

- ΔH_{mix}: Multicomponent alloy enthalpy of mixing from Miedema's model [$\mathbf{kJ \cdot mol^{-1}}$].
- Ω_{ij}: Binary mixing enthalpy at equiatomic composition between elements i and j [$\mathbf{kJ \cdot mol^{-1}}$].
- ΔS_{mix}: Molar configurational entropy of mixing [$\mathbf{J \cdot mol^{-1} \cdot K^{-1}}$]. High-entropy criterion: $\Delta S_{\mathrm{mix}} \geq 1.5R \approx 12.47\,\mathbf{J \cdot mol^{-1} \cdot K^{-1}}$.
- c_i, c_j: Atomic fractions of components i and j [dimensionless].

2.2 Atomic Size Mismatch Parameter (δ) & Electronegativity Difference (Δχ)

Where Used: HEAThermoEngine.m. Evaluates Hume-Rothery topological strain ($\delta \leq 6.6\%$) and chemical bonding polarity.

$$\delta = \sqrt{\sum_{i=1}^{n} c_i \left(1 - \frac{r_i}{\bar{r}}\right)^2} \times 100\%, \quad \bar{r} = \sum_{i=1}^{n} c_i r_i \tag{9}$$

$$\Delta\chi = \sqrt{\sum_{i=1}^{n} c_i (\chi_i - \bar{\chi})^2}, \quad \bar{\chi} = \sum_{i=1}^{n} c_i \chi_i \tag{10}$$

Variable Definitions:

- δ: Atomic size difference mismatch parameter [$\%$]. Solid solution stability requires $\delta \leq 6.6\%$.
- r_i: Empirical Goldschmidt atomic radius of element i [$\mathbf{\AA}$].
- $\bar{r}$: Average atomic radius of the alloy system [$\mathbf{\AA}$].
- χ_i: Pauling electronegativity of pure element i [dimensionless].
- $\Delta\chi$: Root-mean-square electronegativity difference across the alloy matrix.

2.3 Yang-Zhang Thermodynamic Ω Parameter

Where Used: HEAThermoEngine.m. Ratio of entropic stabilization to mixing enthalpy for solid solution formation.

$$\Omega = \frac{T_m \, \Delta S_{\text{mix}}}{|\Delta H_{\text{mix}}|}, \quad T_m = \sum_{i=1}^{n} c_i T_{m,i} \tag{11}$$

Variable Definitions:

- Ω: Dimensionless Yang-Zhang stability parameter. Solid solution formation is favored when $\Omega \geq 1.1$ and $\delta \leq 6.6\%$.
- T_m: Rule-of-mixtures theoretical melting point of the alloy [$\mathbf{K}$].
- $T_{m,i}$: Melting point of pure constituent element i [$\mathbf{K}$].

2.4 Guo Valence Electron Concentration (VEC) Phase Stability

Where Used: HEAThermoEngine.m. Predicts crystal structure: $\mathbf{VEC} \geq 8.0 \implies \mathbf{FCC}$; $\mathbf{VEC} < 6.87 \implies \mathbf{BCC}$; $6.87 \leq \mathbf{VEC} < 8.0 \implies \mathbf{Duplex}$.

$$\mathbf{VEC} = \sum_{i=1}^{n} c_i \mathbf{VEC}_i \tag{12}$$

Variable Definitions:

- $\mathbf{VEC}$: Total valence electron concentration [electrons/atom].
- $\mathbf{VEC}_i$: Valence electron count of element i including valence d electrons (Al: 3, Ti: 4, V: 5, Cr: 6, Mn: 7, Fe: 8, Co: 9, Ni: 10, Cu: 11).

2.5 Scheil-Gulliver Non-Equilibrium Solidification & Homogenization Kinetics

Where Used: HEAThermoEngine.m. Calculates non-equilibrium freezing window, segregation severity, and homogenization soaking conditions.

$$T_{\text{ScheilSolidus}} = T_{\text{Solidus}} - 0.45 \cdot (T_{\text{Liquidus}} - T_{\text{Solidus}}) \cdot (1 + 0.05 \cdot \delta) \tag{13}$$

$$I_{\text{seg}} = 0.25 + 0.06 \cdot \delta, \quad T_{\text{homog}} = \min(T_{\text{Solidus}} - 50\ \text{K},\ 0.88 \cdot T_{\text{Solidus}}), \quad \tau_{\text{homog}} = 10.0 \cdot \left(\frac{I_{\text{seg}}}{0.35} \right)^2 \text{hours} \tag{14}$$

Variable Definitions:

- $T_{\text{ScheilSolidus}}$: Non-equilibrium Scheil solidus temperature accounting for solute rejection [$\mathbf{K}$].
- $T_{\text{Solidus}}, T_{\text{Liquidus}}$: Equilibrium solidus and liquidus temperatures [$\mathbf{K}$].
- I_{seg}: Microsegregation severity index.
- T_{homog}: Safe thermal homogenization annealing temperature [$\mathbf{K}$].
- τ_{homog}: Required isothermal soaking time to eliminate dendritic coring [$\mathbf{hours}$].

3. Atomic Lattice Distortion & Soft-Sphere Physics

Primary Engine: HEALatticeDistortionEngine.m (He & Yang, 2018) | Workstation Modules: Tab 4, Tab 6, Diagnostics Console

3.1 Effective Relaxed Atomic Radius

Where Used: HEALatticeDistortionEngine.m. Soft-sphere elastic atomic relaxation in concentrated multicomponent alloys.

$$r_i^* = r_i \left(1 - \beta \frac{K_i - \bar{K}}{\bar{K}} \right), \quad \beta = \frac{1 + \nu}{3(1 - \nu)}, \quad \bar{K} = \sum_{i=1}^{n} c_i K_i \tag{15}$$

Variable Definitions:

- r_i^*: Effective relaxed atomic radius of element i in the random solid solution [$\mathbf{\AA}$].
- r_i: Unperturbed Goldschmidt neutral radius [$\mathbf{\AA}$].
- $K_i, \bar{K}$: Bulk modulus of pure element i and average alloy bulk modulus [$\mathbf{GPa}$].
- ν: Average alloy Poisson's ratio [dimensionless, $\approx 0.30 - 0.33$].
- β: Elastic stress accommodation factor.

3.2 Root-Mean-Square (RMS) Atomic Lattice Strain

Where Used: HEALatticeDistortionEngine.m. Quantifies intrinsic static lattice displacement strain and dislocation scattering.

$$\varepsilon_{\text{RMS}} = \sqrt{\sum_{i=1}^{n} c_i \left(\frac{r_i^* - \bar{r}^*}{\bar{r}^*} \right)^2}, \quad \bar{r}^* = \sum_{i=1}^{n} c_i r_i^* \tag{16}$$

Variable Definitions:

- ε_{RMS}: Volume-averaged Root-Mean-Square atomic static displacement strain [dimensionless].

- $\bar{r}^*$: Mean effective atomic radius of the relaxed multicomponent lattice [$\mathbf{\AA}$].

3.3 Theoretical Critical Shear Strain

Where Used: HEALatticeDistortionEngine.m. Establishes theoretical plastic shear strain limit of the distorted crystalline lattice.

$$\bar{\gamma}_{\text{th}} = \gamma_0 \left(1 - \alpha_{\text{dist}} \varepsilon_{\text{RMS}}\right) \tag{17}$$

Variable Definitions:

- $\bar{\gamma}_{\text{th}}$: Theoretical plastic shear strain limit of the distorted crystalline lattice.
- γ_0: Ideal theoretical shear strain of an unperturbed pure element crystal ($\gamma_0 \approx 0.11$).
- α_{dist}: Empirical lattice distortion coupling coefficient ($\alpha_{\text{dist}} \approx 2.8$).

3.4 Labusch Collective Pinning vs Peierls Friction Slip

Where Used: HEALatticeDistortionEngine.m. Dislocation mobility transition between discrete Peierls glide and collective solute pinning.

$$\tau_c = \tau_{\text{Peierls}} + \Delta\tau_{\text{Labusch}}, \quad \Delta\tau_{\text{Labusch}} = A_L \cdot G \cdot \varepsilon_{\text{RMS}}^{4/3} \cdot c_i^{2/3} \tag{18}$$

Variable Definitions:

- τ_c: Critical resolved shear stress for dislocation slip [$\mathbf{MPa}$].
- τ_{Peierls}: Intrinsic lattice resistance to dislocation motion [$\mathbf{MPa}$].
- $\Delta\tau_{\text{Labusch}}$: Solid solution strengthening increment from collective pinning [$\mathbf{MPa}$].
- G: Isotropic shear modulus of the alloy [$\mathbf{GPa}$].
- A_L: Dimensionless Labusch scaling coefficient.

3.5 Dislocation Activation Volume

Where Used: HEALatticeDistortionEngine.m. Thermally activated dislocation glide and obstacle bypassing volume.

$$V^* = b \cdot d_w \cdot l_s = -\frac{\partial \Delta G^*}{\partial \tau} \tag{19}$$

Variable Definitions:

- V^*: Dislocation activation volume [$\mathbf{m^3}$ or in units of b^3].
- b: Magnitude of Burgers vector [$\mathbf{nm}$] (≈ 0.255 $\mathbf{nm}$ for FCC, ≈ 0.286 $\mathbf{nm}$ for BCC).
- d_w: Obstacle interaction barrier width [$\mathbf{nm}$].
- l_s: Average spacing between collective solute obstacles [$\mathbf{nm}$].

4. Mechanical Strengthening & Constitutive Formulations

Primary Engine: HEAMechanicalEngine.m | Workstation Modules: Tab 4, Tab 7, Tab 8

4.1 Total Yield Strength Superposition

Where Used: HEAMechanicalEngine.m. Superposition model for room-temperature engineering tensile yield strength.

$$\sigma_y = \sigma_0 + \Delta\sigma_{\text{ss}} + \Delta\sigma_{\text{HP}} + \Delta\sigma_{\text{ppt}} \tag{20}$$

Variable Definitions:

- σ_y: Room-temperature engineering tensile yield strength (0.2% offset) [$\mathbf{MPa}$].
- σ_0: Intrinsic lattice friction stress [$\mathbf{MPa}$] (~ 45 $\mathbf{MPa}$ for FCC; $\sim 650 - 950$ $\mathbf{MPa}$ for BCC).
- $\Delta\sigma_{\text{ss}}$: Solid solution strengthening contribution from atomic size and modulus mismatch [$\mathbf{MPa}$].
- $\Delta\sigma_{\text{HP}}$: Hall-Petch grain boundary strengthening [$\mathbf{MPa}$].
- $\Delta\sigma_{\text{ppt}}$: Precipitation / intermetallic dispersion strengthening [$\mathbf{MPa}$].

4.2 Varvenne-Curtin (FCC) & Maresca-Curtin (BCC) Solid Solution Models

Where Used: HEAMechanicalEngine.m. Structure-specific physical solid solution strengthening models.

$$\Delta\sigma_{\text{ss}}^{\text{FCC}} = 0.0405 \cdot M \cdot \mu \left(\frac{1+\nu}{1-\nu}\right)^{4/3} \left[\sum_{i=1}^{n} c_i \left(\frac{\Delta V_i}{V}\right)^2\right]^{2/3} \tag{21}$$

$$\Delta\sigma_{\text{ss}}^{\text{BCC}} = M \cdot \left[\tau_{\text{kink}} + \alpha_k \cdot \mu \cdot \Delta\varepsilon_{\text{size}}^{4/3}\right] \tag{22}$$

Variable Definitions:

- M: Taylor orientation factor ($M = 3.06$ for FCC; $M = 2.73$ for BCC).

- μ: Effective isotropic shear modulus of the alloy [$\mathbf{GPa}$].
- ν: Poisson's ratio [dimensionless, typically $\mathbf{0.30 - 0.33}$].
- $\Delta V_i/V$: Normalized misfit volume of solute atom i relative to average atomic volume V.
- τ_{kink}: Solute-resisted kink-pair nucleation threshold along screw dislocation lines [$\mathbf{MPa}$].

4.3 Hall-Petch Strengthening & High-Temperature Softening

Where Used: HEAMechanicalEngine.m. Grain boundary hardening and high-temperature yield strength decay.

$$\Delta\sigma_{\text{HP}} = \frac{k_{\text{HP}}}{\sqrt{d}}, \quad \sigma_y(T) = \sigma_y(298\ \text{K})\left[1 - \left(\frac{T - 298}{T_m - 298}\right)^m\right] \tag{23}$$

Variable Definitions:

- k_{HP}: Hall-Petch grain boundary strengthening coefficient [$\mathbf{MPa \cdot \mu m^{1/2}}$]. ($\mathbf{494\,MPa \cdot \mu m^{1/2}}$ for Cantor).
- d: Mean grain diameter [$\mu\mathbf{m}$]. Standard reference annealed state: $d = 30\,\mu\mathbf{m}$.
- T_m: Solidus or melting temperature of the alloy [$\mathbf{K}$].
- m: Empirical thermal softening exponent ($m \approx \mathbf{0.65 - 0.85}$).

4.4 Hardness, Flow Stress, Ductility & Wear

Where Used: HEAMechanicalEngine.m. Vickers hardness, Hollomon plastic flow curve, Archard wear loss, and cladding dilution.

$$\text{HV} \approx 0.3\,\sigma_y + 60, \quad \sigma_{\text{true}} = K\,\varepsilon_{\text{true}}^n, \quad W_{\text{wear}} = K_{\text{Archard}}\frac{F_N\,s}{H}, \quad \eta = \frac{A_{\text{sub}}}{A_{\text{clad}} + A_{\text{sub}}} \times 100\% \tag{24}$$

Variable Definitions:

- $\mathbf{HV}$: Vickers microhardness number [$\mathbf{kgf \cdot mm^{-2}}$].
- K: Strength coefficient in Hollomon constitutive law [$\mathbf{MPa}$].
- n: Strain-hardening exponent (typically $\mathbf{0.35 - 0.50}$ for TWIP/FCC; $\mathbf{0.08 - 0.15}$ for BCC).
- B/G: Pugh ratio of bulk modulus B to shear modulus G. Ductile regime: $B/G > \mathbf{1.75}$.
- W_{wear}: Total volumetric wear loss [$\mathbf{mm^3}$].
- K_{Archard}: Dimensionless Archard wear coefficient.
- F_N: Normal contact load applied to wear interface [$\mathbf{N}$].
- s: Total sliding distance [$\mathbf{m}$].
- η: Laser cladding substrate dilution percentage [$\%$].

5. Thermophysical Transport Properties

Primary Engine: HEATransportEngine.m | Workstation Modules: Tab 3 (Transport Curves)

5.1 Density, Heat Capacity & Thermal Conductivity

Where Used: HEATransportEngine.m. Temperature-dependent density, specific heat capacity, and thermal conductivity.

$$\rho(T) = \frac{\rho(298\ \text{K})}{1 + 3\,\alpha_L(T - 298)}, \quad C_p(T) = \sum_{i=1}^{n} c_i\,C_{p,i}(T), \quad k(T) = \frac{L_0\,T}{\rho_{\text{el}}(T)} + \frac{A_{\text{ph}}}{T} \tag{25}$$

Variable Definitions:

- $\rho(T)$: Volumetric mass density at temperature T [$\mathbf{kg \cdot m^{-3}}$ or $\mathbf{g \cdot cm^{-3}}$].
- α_L: Mean linear coefficient of thermal expansion [$\mathbf{K^{-1}}$].
- $C_p(T)$: Molar specific heat capacity at constant pressure [$\mathbf{J \cdot mol^{-1} \cdot K^{-1}}$].
- $C_{p,i}(T)$: Specific heat of pure constituent i derived from SGTE unary data.
- $k(T)$: Total thermal conductivity [$\mathbf{W \cdot m^{-1} \cdot K^{-1}}$].
- L_0: Sommerfeld theoretical Lorenz number ($\mathbf{2.443 \times 10^{-8}\,W \cdot \Omega \cdot K^{-2}}$).
- $\rho_{\text{el}}(T)$: Temperature-dependent electrical resistivity [$\mathbf{\Omega \cdot m}$].

5.2 Semi-Solid Rheology & Laser Absorptivity

Where Used: HEATransportEngine.m. Slurry apparent viscosity during freezing and laser powder bed optical absorptivity.

$$\eta(f_s) = \eta_{\text{liq}}\left(1 - \frac{f_s}{f_{s,\text{crit}}}\right)^{-2.5\,f_{s,\text{crit}}}, \quad A_{\text{laser}}(\lambda, T) \approx \sqrt{\frac{8\pi\varepsilon_0 c_0\,\rho_{\text{el}}(T)}{\lambda}} \tag{26}$$

Variable Definitions:

- $\eta(f_s)$: Dynamic apparent shear viscosity of semi-solid slurry in mushy freezing zone [$\mathbf{Pa \cdot s}$].

- η_{liq}: Viscosity of fully molten liquid alloy ($\eta_{\text{liq}} \approx 0.003 - 0.006\,\text{Pa}\cdot\text{s}$).
- $f_{s,\text{crit}}$: Critical solid fraction at rheological percolation jamming ($f_{s,\text{crit}} \approx 0.50 - 0.60$).
- A_{laser}: Dimensionless optical surface absorptivity under industrial laser radiation.
- ε_0: Vacuum permittivity ($8.854 \times 10^{-12}\,\text{F}\cdot\text{m}^{-1}$).
- c_0: Speed of light in vacuum ($2.9979 \times 10^{8}\,\text{m}\cdot\text{s}^{-1}$).

6. Environmental Durability: Oxidation & Corrosion Kinetics

Primary Engines: HEAOxidationEngine.m, HECorrosionEngine.m | Workstation Modules: Tab 5, Tab 8

6.1 Wagner High-Temperature Parabolic Oxidation Law

Where Used: HEAOxidationEngine.m. Isothermal high-temperature oxidation mass gain and Ellingham oxide stability hierarchy.

$$\left(\frac{\Delta W}{A}\right)^2 = k_p(T)\,t, \quad k_p(T) = k_0 \exp\left(-\frac{Q_{\text{ox}}}{RT}\right), \quad \text{PBR} = \frac{M_{\text{ox}}\,\rho_{\text{met}}}{n_{\text{met}}\,M_{\text{met}}\,\rho_{\text{ox}}} \tag{27}$$

Variable Definitions:

- $\Delta W/A$: Specific mass gain per unit exposed surface area [$\text{mg}\cdot\text{cm}^{-2}$].
- $k_p(T)$: Parabolic oxidation rate constant [$\text{mg}^2\cdot\text{cm}^{-4}\cdot\text{h}^{-1}$].
- t: Isothermal exposure duration [hours].
- Q_{ox}: Apparent activation energy for scale growth [$\text{kJ}\cdot\text{mol}^{-1}$].
- PBR: Pilling-Bedworth Ratio. Protective scale criterion: $1.0 \le \text{PBR} \le 2.0$.
- $\Delta G_f^\circ(T)$: Standard formation Gibbs energy per mole of O_2 [$\text{kJ}\cdot\text{mol}^{-1}$].

6.2 Electrochemical Corrosion Rate (ASTM G102), PREN & Butler-Volmer

Where Used: HECorrosionEngine.m. Electrochemical corrosion penetration rate and micro-galvanic cell polarization.

$$\text{CR} = K_{\text{corr}}\,\frac{i_{\text{corr}}\,\text{EW}}{\rho}, \quad \text{EW} = \left[\sum_{i=1}^{n}\frac{w_i\,n_i}{M_i}\right]^{-1}, \quad \text{PREN}_{\text{HEA}} = \text{wt\%Cr} + 3.3(\text{wt\%Mo} + 0.5\,\text{wt\%W}) + 16\,\text{wt\%N} \tag{28}$$

$$i_{\text{net}} = i_{\text{corr}}\left[\exp\left(\frac{2.303(E - E_{\text{corr}})}{\beta_a}\right) - \exp\left(-\frac{2.303(E - E_{\text{corr}})}{\beta_c}\right)\right] \tag{29}$$

Variable Definitions:

- CR: Annual corrosion penetration rate [$\text{mm}\cdot\text{year}^{-1}$].
- K_{corr}: Metric conversion constant ($0.00327\,\text{mm}\cdot\text{g}\cdot\mu\text{A}^{-1}\cdot\text{cm}^{-1}\cdot\text{year}^{-1}$).
- i_{corr}: Corrosion current density determined from Tafel extrapolation [$\mu\text{A}\cdot\text{cm}^{-2}$].
- EW: Alloy equivalent weight [$\text{g}\cdot\text{equivalent}^{-1}$].
- w_i: Mass fraction of element i.
- n_i: Valence electron exchange in dissolution reaction (e.g. Fe: 2, Cr: 3, Ni: 2, Mo: 4).
- PREN_{HEA}: Pitting Resistance Equivalent Number. Pitting immunity in seawater requires $\text{PREN} \ge 40$.
- β_a, β_c: Anodic and cathodic Tafel slopes [$\text{V}\cdot\text{decade}^{-1}$].

7. Kinematic Powder X-Ray Diffractometry

Primary Engine: HEAXRDEngine.m | Workstation Modules: Tab 6 (Diffractogram, Peak Broadening Slider)

7.1 Bragg's Law, Interplanar Spacing & Peak Intensity

Where Used: HEAXRDEngine.m. Synthesizes theoretical powder X-ray diffractogram patterns for FCC, BCC, and HCP lattices.

$$\lambda = 2\,d_{hkl}\,\sin\theta_{hkl}, \quad d_{hkl} = \frac{a_0}{\sqrt{h^2 + k^2 + l^2}} \tag{30}$$

$$I_{hkl} \propto |F_{hkl}|^2 \cdot p_{hkl} \cdot \left(\frac{1 + \cos^2(2\theta)}{\sin^2\theta\cos\theta}\right) \cdot \exp(-2B_{\text{DW}}\sin^2\theta/\lambda^2) \tag{31}$$

Variable Definitions:

- λ: X-ray wavelength ($\lambda = 1.54056\,\text{\AA}$ for Cu-Kα_1).
- θ_{hkl}: Bragg diffraction angle [degrees].
- d_{hkl}: Interplanar spacing of (hkl) lattice planes [$\text{\AA}$].
- a_0: Effective lattice parameter calculated from Vegard's law with soft-sphere relaxation [$\text{\AA}$].
- F_{hkl}: Kinematic crystal structure factor for (hkl) reflection.

- p_{hkl}: Multiplicity factor of (hkl) plane family.
- B_{DW}: Debye-Waller factor for static displacement and thermal vibration [$\mathbf{\AA^2}$].

7.2 Scherrer Peak Broadening & Williamson-Hall Microstrain

Where Used: HEAXRDEngine.m. Simulates crystallite size and lattice microstrain peak broadening.

$$\mathrm{FWHM}_{hkl} = \frac{K_{\mathrm{Scherrer}}\,\lambda}{D_{\mathrm{crystallite}}\cos\theta_{hkl}} + 4\,\varepsilon_{\mathrm{micro}}\tan\theta_{hkl} \tag{32}$$

Variable Definitions:

- FWHM_{hkl}: Diffraction peak Full Width at Half Maximum [$\mathbf{rad}$].
- K_{Scherrer}: Crystallite shape factor ($K_{\mathrm{Scherrer}} \approx 0.94$).
- $D_{\mathrm{crystallite}}$: Coherent crystalline domain size [$\mathbf{nm}$].
- $\varepsilon_{\mathrm{micro}}$: Root-mean-square microstrain from lattice distortion.

8. Industrial Application Ranking & COMSOL Coupling

Primary Engines: HEAApplicationDatabase.m, CALPHADHEACalc.m | Workstation Modules: Tab 7, Tab 8

8.1 Multi-Criteria Decision Analysis (MCDA) Scoring

Where Used: HEAApplicationDatabase.m. Multi-criteria suitability ranking across 8 industrial sectors.

$$S_j = \left[\sum_{k=1}^{m} w_k \cdot \phi_k(P_k)\right] \times \prod_{l=1}^{p} \Theta_l, \quad \phi_k(P_k) = \frac{1}{1 + \exp\left(-\kappa_k \cdot \frac{P_k - P_{k,\mathrm{target}}}{\Delta P_k}\right)} \tag{33}$$

Variable Definitions:

- S_j: Overall industrial suitability score for application category j ($0 \le S_j \le 1.0$).
- w_k: Relative importance weighting factor assigned to property k ($\sum w_k = 1.0$).
- $\phi_k(P_k)$: Continuous non-linear sigmoid response utility function evaluating property $P_k \in [0,1]$.
- $\Theta_l \in \{0,1\}$: Critical hard failure constraint gate (e.g. elongation $< 5\%$ disqualifies cryogenic LNG applications).

8.2 COMSOL Multiphysics Piecewise Continuous Smoothing

Where Used: CALPHADHEACalc.m. Exports differentiable material property functions for COMSOL finite element simulations.

$$P_{\mathrm{smooth}}(T) = P_{\mathrm{solid}}(T) + \left[P_{\mathrm{liquid}}(T) - P_{\mathrm{solid}}(T)\right] \cdot \mathrm{flc2hs}\left(T - T_{\mathrm{Solidus}},\, \Delta T_{\mathrm{smooth}}\right) \tag{34}$$

Variable Definitions:

- $P_{\mathrm{smooth}}(T)$: Continuous, differentiable material property function valid across solid, mushy, and liquid regimes.
- $\mathrm{flc2hs}(x, \Delta x)$: COMSOL continuous Heaviside step operator with continuous second derivative.
- $\Delta T_{\mathrm{smooth}}$: Smoothing transition temperature band [$\mathbf{K}$].

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